

Custom Sheet Metal Pallet AMR ($1000\text{ mm} \times 1200\text{ mm}$) - Master CAD Design & Dimensioning Guide

This document presents a complete mechanical analysis, rough CAD part dimensions, and sheet metal fabrication guidelines for your custom $1000\text{ mm} \times 1200\text{ mm}$ **Low-Profile Pallet AMR** based on the 3 hand-drawn notebook design sketches (Pages 8, 11, and 12).

1. Design Concept & Kinematic Mechanism Analysis

Working Principle (Based on Notebook Sketches)

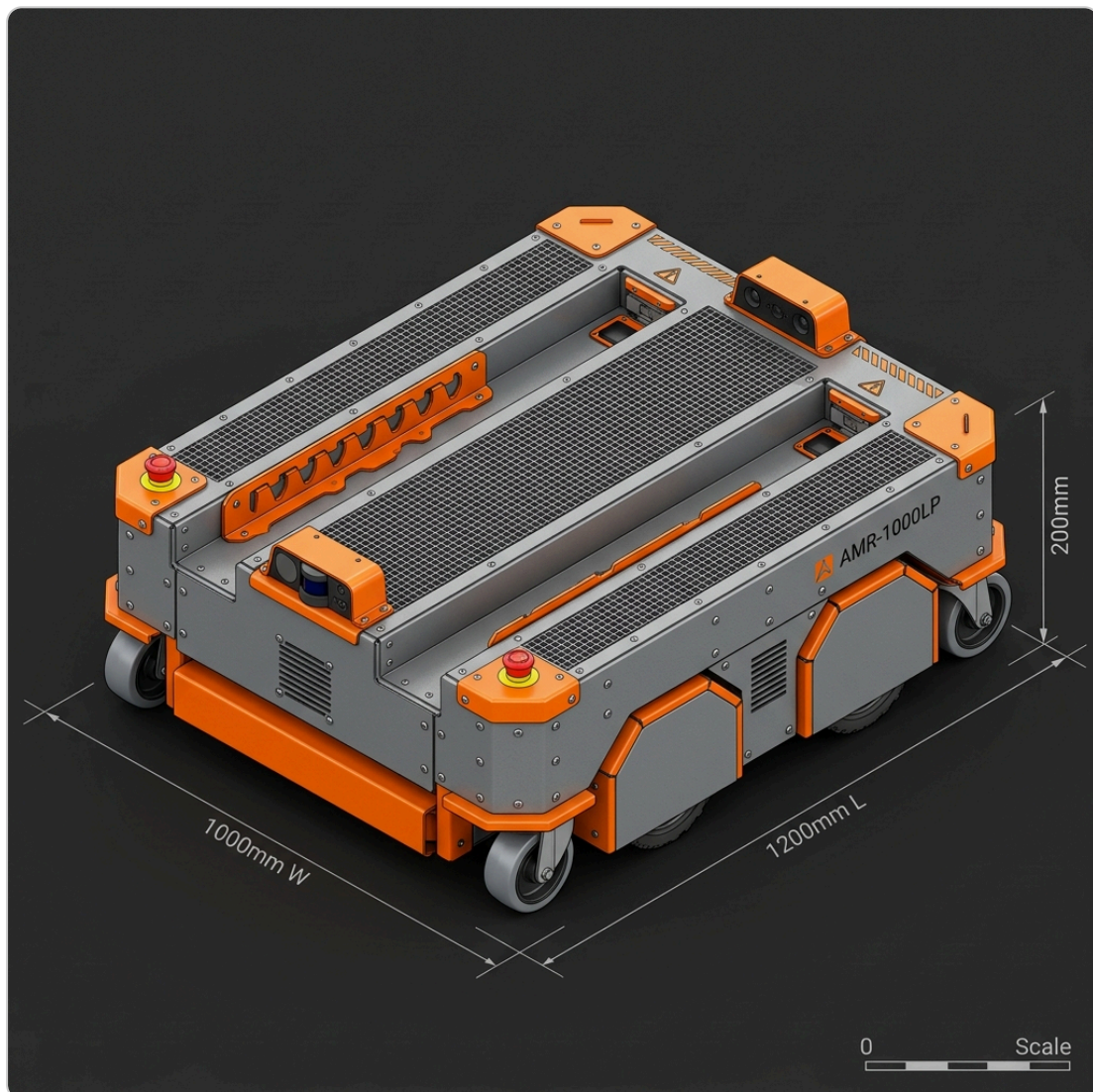
As shown in your hand-drawn sketches:

- **Notebook Page 8 (April 8):** Shows the isometric top view of the $1000\text{ mm} \times 1200\text{ mm}$ AMR body with two parallel recessed fork channels and 10 guide follower pin positions along the track.
- **Notebook Page 11 (April 11):** Details the **Variable Linear Guide Height / Z-Ramp Cam Slot Mechanism** (Ramp profiles 1 through 10).
- *Why this works:* When the pallet forks retract inside the main chassis body, the cam follower pins ride UP the sloped Z-ramp slot. This **elevates the fork carriage vertically by 60 mm** , lifting the pallet's bottom load wheels off the floor so the main AMR chassis can rotate and navigate freely!
- **Notebook Page 12 (April 12):** Shows the alternative scissor/pantograph linkage and the lower drive belt/wheel assembly layout.

2. 3D CAD Visualization & Master Blueprint

A. 3D CAD Concept Render ($1000\text{ mm W} \times 1200\text{ mm L} \times 200\text{ mm H}$)

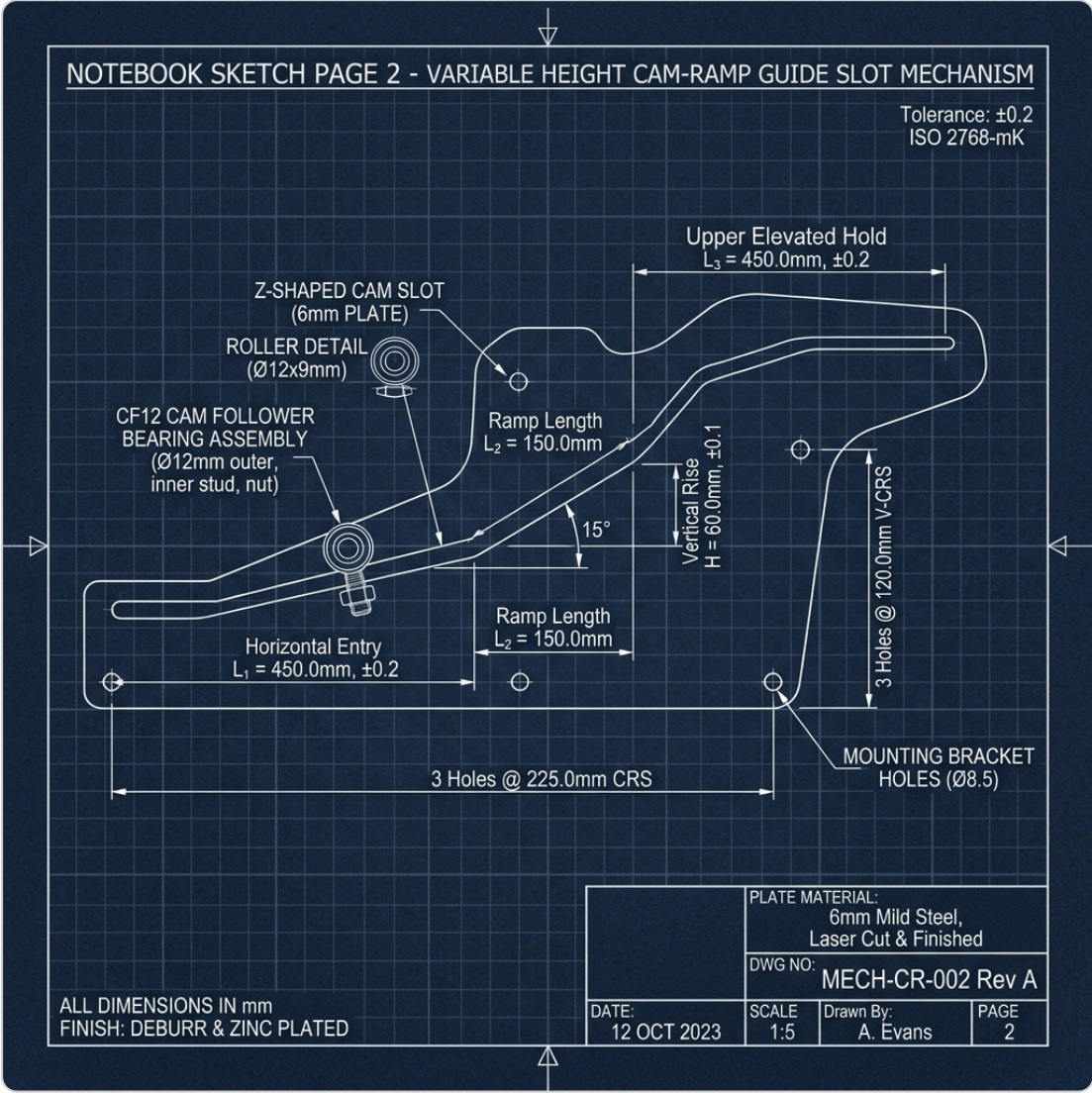
Constructed with bended 3.0 mm steel sheet metal, side drive compartments, laser-cut cam guide slot side plates, corner casters, and two recessed fork channels:



Custom 1000mm x 1200mm Sheet Metal Pallet AMR 3D CAD Concept Render (Illustrative Schematic / Non-Realistic)

B. 2D Engineering Blueprint - Variable Height Cam-Ramp Slot Mechanism

Detailing the Z-shaped sloped cam slot track profile drawn on Notebook Page 11:



Notebook Page 11 Variable Height Cam-Ramp Guide Slot Mechanism 2D Blueprint (Illustrative Schematic / Non-Realistic)

3. Recommended Overall Height & Footprint Sizing

Parameter	Recommended Dimension	Design Rationale & Clearance
Overall Width (W)	1000 mm	User target width
Overall Length (L)	1200 mm	User target length (fits standard 1200 mm ISO / EUR pallets)
Lowered Chassis Height (H_{low})	200 mm	Calculated Minimum Height: Includes 160 mm drive wheel, 25 mm ground clearance, and 3.0 mm top/bottom sheet metal deck plates
Raised Deck Height (H_{raised})	260 mm	Adds 60 mm vertical elevation travel from the cam Z-ramp
Pallet Entry Fork Height	90 mm	Lowered fork thickness to enter standard pallet pockets (100 mm opening)
Minimum Ground Clearance	25 mm	Base sheet metal plate clearance above factory floor

4. Rough Design Dimensions for ALL Sheet Metal Parts

Use the dimensional table below to begin modeling individual parts and sheet metal features in your CAD software (SolidWorks, Fusion 360, or Inventor):

Part Name / Module	Material & Thickness	Key Dimensions (mm)	CAD Feature & Sheet Metal Notes
1. Main Outer Chassis Hull	3.0\text{ mm} CRCA Mild Steel	1000\text{ W} \times 1200\text{ L} \times 200\text{ H}	Outer bended enclosure with 15\text{ mm} stiffening lips and corner laser cutouts
2. Left & Right Drive Bays	3.0\text{ mm} Mild Steel Sheet	220\text{ W} \times 1200\text{ L} \times 200\text{ H}	Two side compartments housing motors, 24V/48V battery, controllers, and LiDARs
3. Recessed Fork Channels	4.0\text{ mm} Mild Steel Sheet	Outer Width: 560 mm Tyne Width: 160 mm Inner Gap: 240 mm	Centered between side bays (220 + 160 + 240 + 160 + 220 = 1000\text{ mm}). Tyne length: 1050\text{ mm}
4. Cam Guide Slot Side Plates	6.0\text{ mm} Heavy Steel Plate	1100\text{ L} \times 160\text{ H} (Laser cut)	2\times side plates with CNC laser-cut 12.2\text{ mm} wide Z-shaped cam tracks
Cam Track Breakdown:			
-- Horizontal Entry (L_1)	Laser-cut Slot	Length: 450 mm, Height: 40\text{ mm}	Fork lowered entry position (90\text{ mm} deck height)
-- Sloped Ramp (L_2)	15^\circ Inclined Slot	Length: 150 mm, Rise: 60 mm	60\text{ mm} vertical elevation ramp curve
-- Elevated Hold (L_3)	Laser-cut Slot	Length: 450 mm, Height: 100\text{ mm}	Retracted raised position (260\text{ mm} deck height)

Part Name / Module	Material & Thickness	Key Dimensions (mm)	CAD Feature & Sheet Metal Notes
5. Cam Follower Bearings	Stud-Type Bearings (CF12)	$\varnothing 30$ Outer, $\varnothing 12$ Stud	4 Heavy-duty stud cam followers mounted to inner fork carriage
6. Inner Lifting Fork Carriage	4.0 Steel Sheet Weldment	550 W $\times$ 950 L $\times$ 120 H	Bent U-frame carrying the top tynes with tab-and-slot welding alignment notches
7. Drive Wheels (Differential)	Polyurethane on Aluminum	$\varnothing 160$ $\times$ 50 mm Width	2 Center differential drive wheels ($\varnothing 25$ mm shaft bore)
8. Corner Swivel Casters	Polyurethane Casters	$\varnothing 75$ Wheel Dia	4 Corner swivel casters (100 mm total mounting height)
9. Drive Motors & Gearbox	BLDC Servo Motors	750W 24V (1:20 Planetary)	2 Motors mounted inside the 220 mm side bays

5. Sheet Metal Fabrication Guidelines for CAD Modeling

1. Bending Parameters:

- Sheet Thickness $t = 3.0\text{ mm}$ $\rightarrow$ Internal Bend Radius $R = 3.0\text{ mm}$ ($K\text{-factor} = 0.44$).
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2. Bend Reliefs:

- Add rectangular or teardrop bend relief cuts ($\text{Width} = 4.0\text{ mm}$, $\text{Depth} = 5.0\text{ mm}$) at all flange intersections to prevent metal tearing during press brake bending.

3. Tab-and-Slot Alignment Notches:

- Incorporate laser-cut $15\text{ mm} \times 4\text{ mm}$ tabs and matching slots on internal structural bulkheads. This allows the sheet metal pieces to self-align during assembly before plug welding.

4. Cam Track Surface Hardening:

- The 6.0 mm laser-cut cam slot side plates should be case-hardened (nitrided or induction hardened) to prevent localized indentation from the $\geq 30\text{ mm}$ steel cam followers under a 1000 kg payload.